# *A pilot study of potassium supplementation in the treatment of hypokalemic patients with rheumatoid arthritis: a randomized, double-blinded, placeb...*

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Patients with rheumatoid arthritis (RA) have been described as having significantly low serum potassium concentrations than that in healthy subjects. We assessed the therapeutic efficacy and tolerability of oral potassium supplement dissolved in grape juice in female hypokalemic patients with active RA. Thirty-two hypokalemic patients with active RA were investigated in a parallel, randomized design. In addition to their usual medication, the control group received placebo and the intervention group received 6000 mg chloride potassium dissolved in grape juice on 28 consecutive days. The primary outcome parameter was the change of pain on a visual analog scale (VAS). The American College of Rheumatology (ACR) percent response criteria and Disease Activity Score 28 (DAS28, 28-joint count) and the European League Against Rheumatism (EULAR) moderate response were assessed. Mean age was 48.6 +/- 6 years. In the potassium group, 43.75% (7/16) of the patients met the criteria of 33% lower pain intensity compared with 6.25% (1/16) in the placebo group ($P < .02$) at day 28. Also, 31.25% (5/16) of the patients in the intervention group achieved moderate responses, according to the EULAR criteria. The corresponding percentage for patients receiving placebo was 6.25% (1/16) ($P < .05$). Potassium supplements appeared to decrease pain intensity. PERSPECTIVE: This article reports a trial evaluating the effect of potassium supplementation in the treatment of pain in hypokalemic patients with rheumatoid arthritis. The elevated serum cortisol and potassium values in the treatment group correlate negatively with patient's assessment of pain intensity, reflecting an anti-pain effect for potassium supplementation.